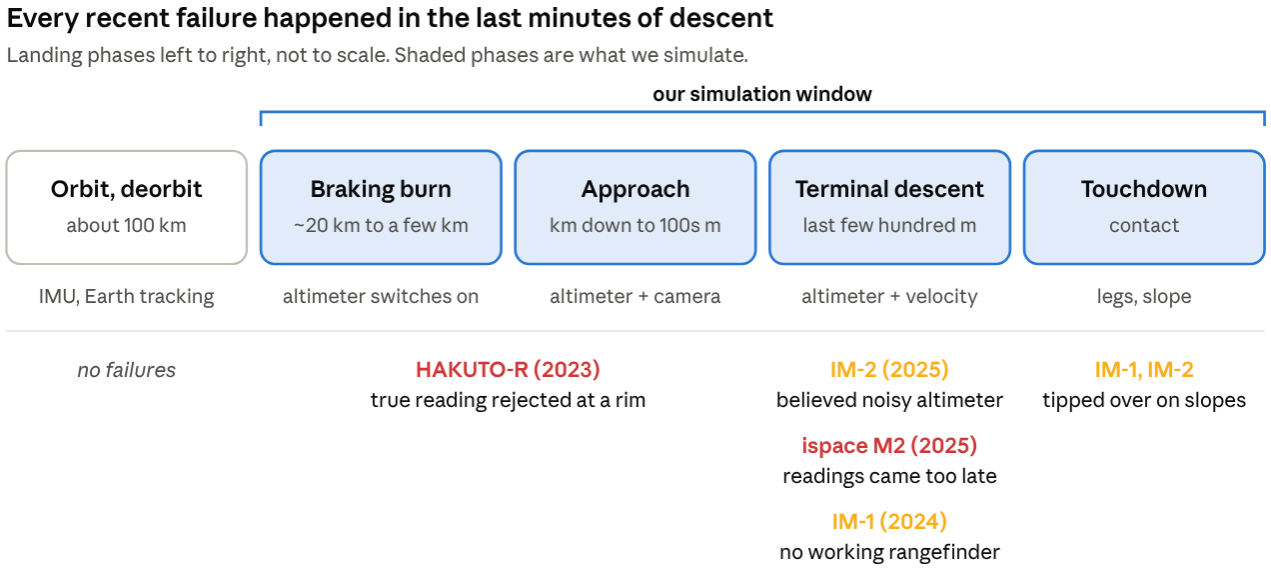
# PRJ1 Objective and Problem Definition: Sensor Trust for Lunar South Pole Landing

We aim to make a lunar lander's navigation filter trust each sensor according to the terrain and lighting ahead, so that a real cliff is not mistaken for a broken sensor and a blinded camera is not believed. We will prove this by landing a simulated lander on real NASA South Pole terrain.

## Problem statement

During powered descent to a lunar South Pole site, a lander's navigation filter must fuse laser altimeter, optical camera and IMU data. It cannot tell a real terrain discontinuity, or a shadowed, featureless scene, from a sensor fault. With a fixed outlier gate and fixed sensor weights, the filter either rejects valid altimeter data at crater rims (the HAKUTO-R M1 failure, April 2023) or keeps trusting a camera that cannot see in shadow (a contributor to IM-2, March 2025).

**Problem:** adjust, ahead of time and along the planned trajectory, how much the filter trusts each sensor, using known terrain and illumination.




landing phases and where recent missions failed

The weakest part of a landing is the handover, when ground-facing sensors first start correcting the lander, and the final descent after it. Our simulation starts there.

## Research questions

1. Can 5 m/pixel LOLA elevation and slope maps predict, before descent, where a true altimeter jump will exceed a standard chi-square innovation gate?
2. Can solar geometry plus the DEM predict where the camera's view will be shadowed enough to make optical navigation unreliable?
3. Does adjusting gate size and sensor weights from these predictions reduce altitude error and false rejections, without letting real sensor faults through?
4. Which way of judging a surprising reading works best: a fixed rule, a map-based tolerance, a "wait and see" consistency check, or a machine-learning classifier trained on simulated descents?
5. Does the trust layer still work at a landing site it was never tuned or trained on, and after a change of site?

## Hypotheses

| # | We expect | How we check |
|---|---|---|
| H1 | The four-step rule rejects far fewer real cliff readings than the fixed rule, without being fooled noticeably more by faults (threshold fixed before testing, for example no more than 1 extra fooled run in 100) | Count both mistakes over about 500 runs |
| H2 | Shadow-aware camera trust reduces position error in dark areas | Error with and without it |
| H3 | When the lander already knows its position well, NASA's 2025 method does as well as ours | "When does it help" sweep over position uncertainty |
| H4 | The rule still works at a site it was never tuned on | Test on Site04 after tuning on Site01 |

H3 predicts where our method should *not* help; confirming it makes the other results more credible.

# Objectives

| # | Objective | Done when |
|---|---|---|
| O1 | **Hazard map:** build per-trajectory "cliff" and "blind-spot" maps for PGDA Site04 (Shackleton rim) and Site01 (connecting ridge) from elevation, slope and height-error maps, with the Sun at 1.5° to 3° | Trust map exported for both sites, including a wide horizon map for shadows |
| O2 | **Trust layer:** turn those maps into per-sensor trust values (gate width and measurement noise) at each point of the descent, plus a check that separates real terrain from sensor faults | Filter runs with the layer switched on or off |
| O3 | **Baseline failures:** reproduce the failure modes of recent missions with a standard fixed-gate filter in a Unity physics simulation: a rejected cliff reading (HAKUTO-R), a shadowed camera with a noisy altimeter (IM-2), delayed readings (ispace M2) and a sensor dropout (IM-1) | Each failure mode reproduced on real terrain |
| O4 | **Evaluation:** compare the baseline, a PITFAL-style filter and the trust layer, including alternative judging methods, over about 500 Monte Carlo runs per scenario | Results table and plots for the metrics below |
| O5 | **Reproducibility and demo:** release the Unity project, code and scenarios, with a side-by-side replay of baseline and trust-layer landings | Others can rerun and watch the results |
| O6 | **Site-change robustness:** compute trust automatically for any site or route, onboard around the current estimated position, and fall back to a cautious mode where no map exists | Trust layer tested on an unseen site and a mid-descent divert without retuning |

**Extension (optional), O7: live 3D scan.** Add a simulated scanning lidar that builds a fresh height patch below the lander, matches it to the stored map to correct position, and joins a three-way check (stored map, live scan, altimeter) to decide which source to trust. Done when landing success with and without the scan is compared on the same descents.

**Metrics for O4:** landing success (% of runs upright within speed and tilt limits); altitude error (RMS, m); false rejection of valid altimeter readings (%); missed injected faults (%); camera weight while in shadow; fuel left at touchdown (kg); landing success on an unseen site compared with the tuned site.

## Scope

| In scope | Out of scope |
|---|---|
| Powered descent from about 15 km altitude to touchdown, including leg contact and tip-over | Orbit, deorbit burn, launch |
| Laser altimeter, optical camera (rendered images), IMU | Building new sensor hardware |
| Physics simulation in Unity over real PGDA DEMs plus procedural terrain | Flight hardware or hardware-in-the-loop tests |
| Navigation filter, trust layer, guidance on estimated state, ML models | Certifying flight software |

Also out of scope: boulder avoidance, engine failures and sensors that never switch on. Plume dust is out of scope for now; the trust layer is built so it can be added later as another condition, following ARC-LIGHT.

**What the project solves, partly solves, and does not solve**

| Solves | Partly solves | Does not solve (stated as limits) |
|---|---|---|
| True readings rejected at cliffs (HAKUTO-R) | Noisy or late readings (IM-2, ispace M2), through probation | Missing or dead sensors (IM-1) |
| Camera trusted in shadow (IM-2) | Site changes, through checks against the map around the current position | Engine failures, boulder avoidance, dust, leg design |

## Assumptions

- The planned trajectory is known before descent.
- The DEM's stated height error is a fair bound on map error.
- Plume dust is a later extension (see ARC-LIGHT in the Literature Review).

## Open questions for the team

- Which Sun azimuths and dates to use for the scenarios.
- Whether to add a velocity sensor such as NDL as a fourth input.